



BRAC UNIVERSITY

CSE360: Computer Interfacing

Lab Project Report

Title: Universal Charging Safety Interlock, Auto-Cutoff, and Welded-Relay Fault Detector

by

Group - 3

Section - 9

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Abstract

This project aims to reduce the risk of battery fires in electric rickshaws by developing a smart battery management system. The system uses different sensors including an INA226 power monitor, DS18B20 temperature sensor and MQ-2 gas sensor in order to continuously monitor the battery condition in real time. An Arduino Uno uses Finite State Machine and Exponential Moving Average (EMA) filters to reduce sensor noise and detect early signs of thermal runaway. The main results show that the system can cut off the power within 2 seconds after detecting a fault. It also automatically stops charging when the battery is fully charged and it saves energy. This improves environmental sustainability by preventing toxic battery fires, reducing e-waste and making battery charging more energy efficient.

Keywords

Battery management system, Thermal Runaway, Computer Interfacing, Sensor Fusion Sustainability, Electric Vehicles

1. Introduction

Electric rickshaws are an important means of transportation in Bangladesh. However, the small garages often face serious fire due to thermal runaway in lithium-ion and lead-acid batteries. This fire is mainly caused by overcharging and the lack of proper temperature monitoring.

The main goal of this project is to develop a low cost and fast response embedded system that continuously monitors battery's voltage, temperature and VOC gas emissions. It also uses software-based data filtering to reduce sensor errors and accurately detect dangerous battery conditions. Overall, the project uses computer interfacing to solve an important real-world safety problem.

2. Related Work/ Inspiration

Commercial Battery Management Systems (BMS) are available in modern and expensive electric vehicles, but they are too costly for local rickshaw garages. Common low-cost chargers usually only check the battery voltage and do not monitor the surrounding environment.

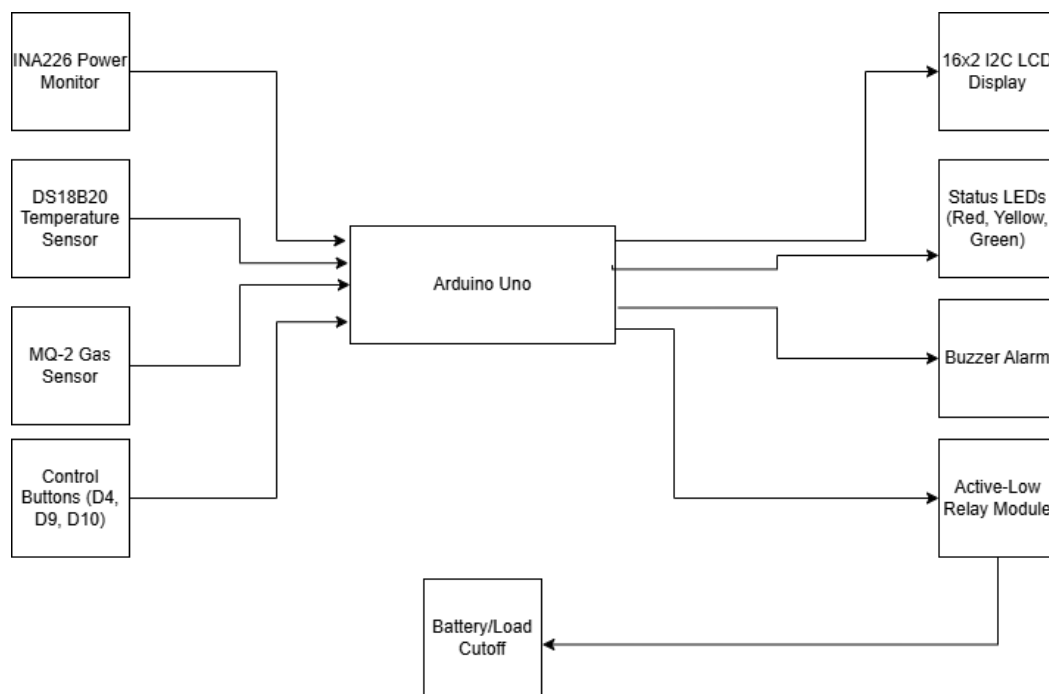
Our project is different because it uses sensor fusion, which combines information from multiple sensors. It checks for both sudden increases in temperature and gas released from the battery, detected by the MQ-2 gas sensor to identify a possible battery problem. This helps the system detect danger early and disconnect the power before a fire starts. It also solves the problem of

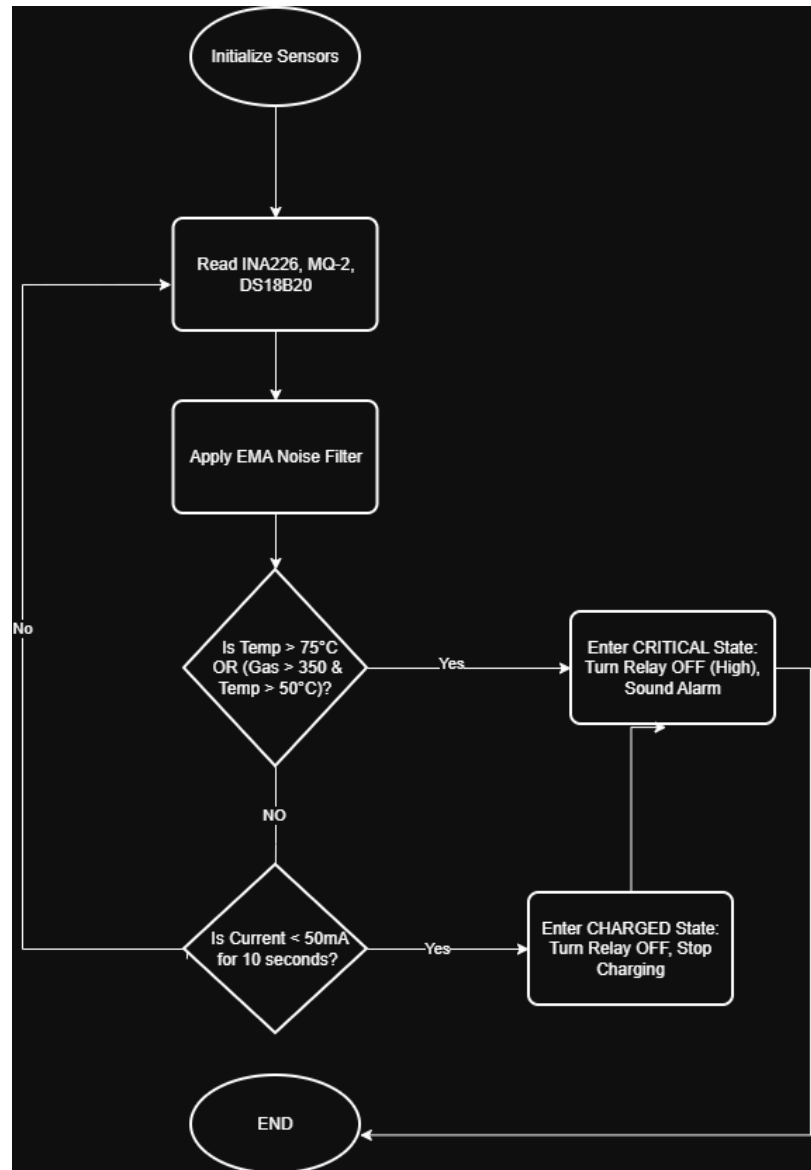
slow response found in standard systems that depend only on the temperature based cut off switches.

3. Technical Approach

System Architecture:

The system uses the main microcontroller to process data from three different sources. A finite state machine(FSM) divides the system into four states: normal, warning, critical and charged. The system uses non-blocking millis() timers to continuously collect sensor data without stopping or freezing the system.





Components Used:

- Microcontroller: Arduino Uno (Central processor handling FSM and logic).
- INA226 (Power Monitor): High-precision voltage and current sensor measuring load.
- DS18B20 (Temperature): Digital thermometer to monitor ambient battery heat.
- MQ-2 (Gas Sensor): Analog sensor to detect VOCs venting from a failing battery.
- Relay Module (Active-Low): Electromechanical switch acting as the fail-safe power cutoff.
- 16x2 I2C LCD & Status LEDs: Provides real-time telemetry and state visualization.

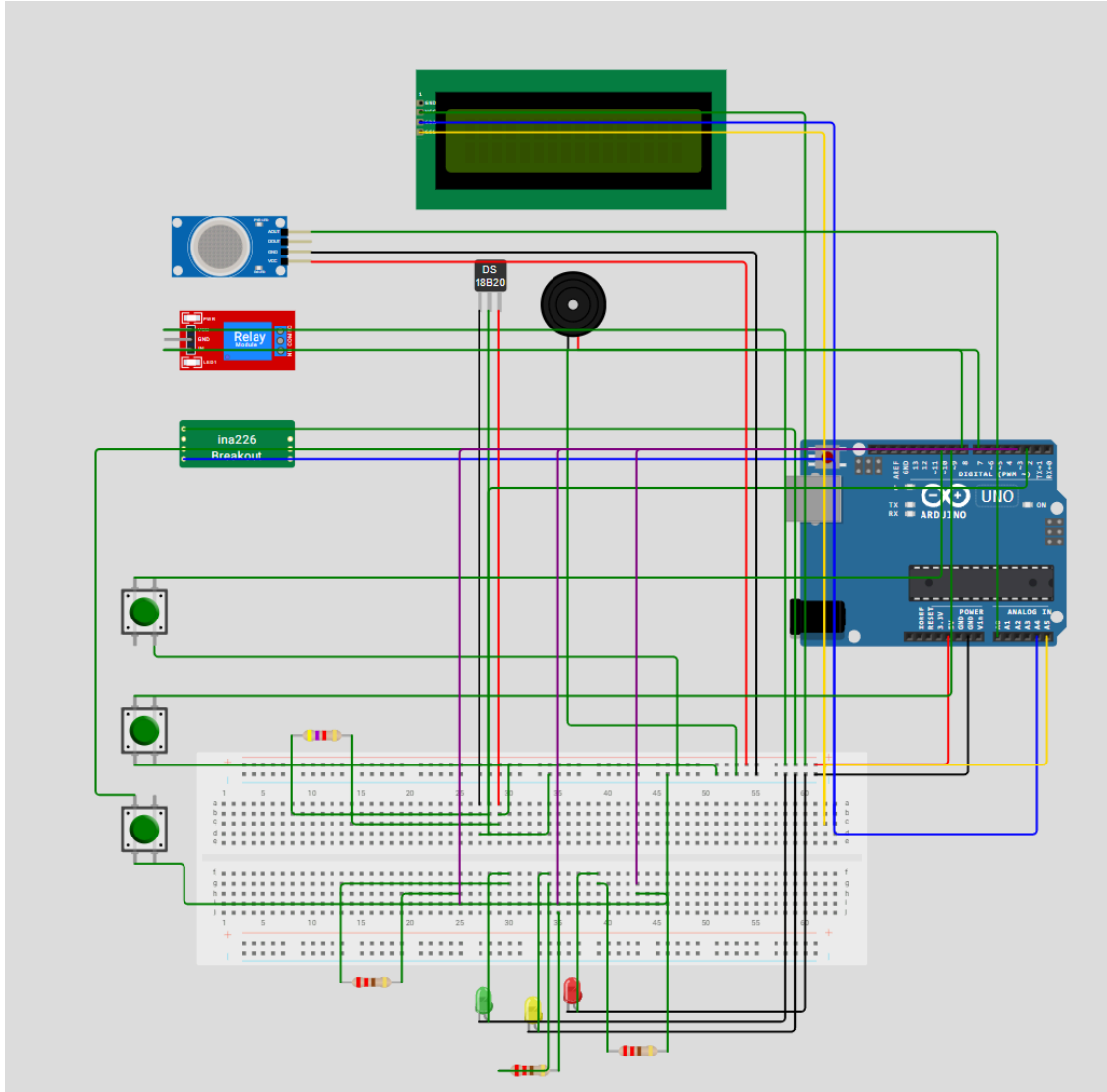
Cost Breakdown:

- Arduino Uno: 940 BDT
- Relay: 80 BDT
- LED - 20 BDT
- Breadboard: 170 BDT
- Jumper wires: 180 BDT
- INA226 Module: 370 BDT
- DS18B20 (Bare Sensor) + Resistor: 350 BDT
- MQ-2 Gas Sensor: 300 BDT
- LCD: 250 BDT
- Cable type C(x2): 400 BDT
- Total Estimated Cost: 3060 BDT

Communication Protocol:

- I2C (Inter-Integrated Circuit): Used for the INA226 and LCD. Justified by its addressable 2-wire architecture, allowing multiple devices on the same bus, saving microcontroller pins.
- 1-Wire Protocol: Used for the DS18B20. Justified by its digital precision and immunity to the voltage drops that plague standard analog thermistors.

Circuit/Schematic:



Implementation:

The system starts by connecting the I2C and 1-wire sensors. The relay is initially set to be low, so charging can begin. Every 300 ms, the system reads data from the sensors. An EMA filter is used to make the reading smoother and remove temporary electrical noise. The system checks two conditions- If temperature is greater than 45 degree Celsius or gas is greater than 350 and temperature is greater than 35 degree celsius or not. If either condition is true the system identifies a possible battery problem. If the dangerous condition continues for 2000 MS the

system changes to the critical state if then changes the relay signal to high, which cuts off the charging power and activates the alarm. This prevents unnecessary charging and saves energy.

Challenges:

- The DS18B20 sensor did not work at first because of a manufacturing dependency. The microchip was soldered in the long direction on the PCB. So, we stopped using the faulty PCB and connected the bare DS18B20 sensor directly using a 4.7 kohm resistor.
- Modern USB-C power banks did not provide power because the CC (Channel Configuration) line was disconnected in our custom cable. So we used a standard 5V / 2A USB adaptor instead and connected the phone side D+/D- pins to bypass the digital handshake and allow the power supply to work.

4. Sustainability & Impact

Sustainability:

This project helps save energy by using a “tail current” monitoring system. It automatically stops charging when the battery is fully charged, preventing unnecessary power consumption or “vampire draw”. It also helps protect the environment by preventing thermal runaway, which can increase the life span of lithium-ion batteries and reduce electronic waste. In addition, the system helps prevent the release of harmful VOC gases and heavy metal smoke into the air that can cause a battery fire.

Impact:

Socially, this system helps protect the lives, safety, livelihoods of low income rickshaw drivers and garage owners. It provides an affordable and reliable safety system that helps prevent serious fires and reduces the risk of major property damage and financial loss.

Future Work:

In the future, the system will be improved by creating a custom PCB to reduce wire resistance. and ESP32 will also be added to provide Wi-Fi and IoT features, allowing the system to send SMS alerts to garage owners. The relay will also be upgraded to safely handle 60V/30A industrial rickshaw chargers.

Limitations:

The current hardware has some limitations. The resistance of the breadboard jumper wires causes voltage drops which limit charging speed during testing. Also, this prototype is designed for 5V USB testing only. To use it with real high-voltage electric vehicle batteries, the system will need stronger physical contractors that can safely handle higher voltage and current.

5. Results & Discussion

The testing results confirm that the FSM system works properly. Under normal conditions, the LCD showed stable readings, such as 4.5 volt to 5.5 volt and changing current of 150-450mA. During the stimulated thermal runaway testing using butane gas, the EMA filter successfully ignored short 1 second gas spikes and prevented false alarms. However, when the gas exposure continues for more than 2000ms, the systems cut off the power every time and change to the critical state. In addition, the load-removal test showed that the system could correctly detect when the battery was fully charged and cut off the power after a 10 second confirmation period This confirmed that the energy saving feature works effectively.

6. Conclusion

The project successfully achieved its goal of developing a multi sensor automatic power cut off system to help prevent battery fires. The main achievement is successfully using I2C and 1-Wire Communication protocols, applying software filters to reduce sensor noise and developing a smooth, non blocking state machine. Overall the prototype shows that advanced safety systems can be developed at a low cost. This system has great potential to improve public safety and environmental sustainability in development areas.

7. Contribution

Name	ID	Role(s) / Responsibilities	Specific Contributions
MD.SAADAT RAHMAN	22201101	Full System Engineering & Implementation, Hardware design, Circuit building, Software design, testing, Report diagrams.	● Hardware: Designed and wired all components. ● Software: Developed functioning. ● Testing: Identified and fixed faulty factory components, adjusted the temperature and gas safety limits. ● Environment setup
Nazah S. Anis	22201105	Documentation of Report and Circuit design, Breadboard connection.	● Documentation: Authored the entire technical report, diagrams, and sustainability impact assessment. ● Breadboard connection ● Environment Setup
shabnam sharmin tarin	22201031	Report	Report
Ehsan Ul Haque Fahim	22201422	Report	Report

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